ECON 470 Homework 4 Submission 1

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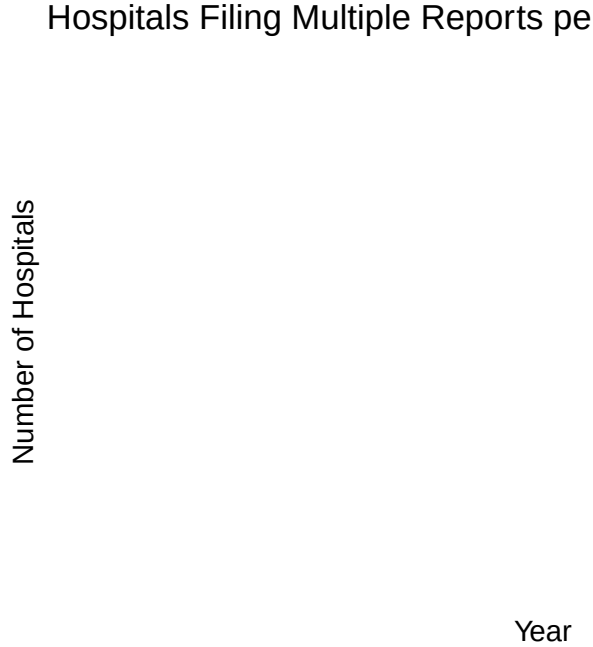
Summarize the Data

Question 1: From 2009-2019, how many hospitals filed more than one report in the same year? Show your answer as a line graph of the number of hospitals over time.



Question 2: After removing/combining multiple reports, how many unique hospital IDs (Medicare provider numbers) exist in the data?

```
# A tibble: 1 x 1
      n
```

```
  <int>
1  6897
```

Question 3: From 2009-2019, what is the distribution of total charges (tot_charges in the data) in each year? Show your results with a "violin" plot, with charges on the y-axis and years on the x-axis. For a nice tutorial on violin plots, look at Violin Plots with ggplot2.

```
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(`stat_ydensity()`).
```
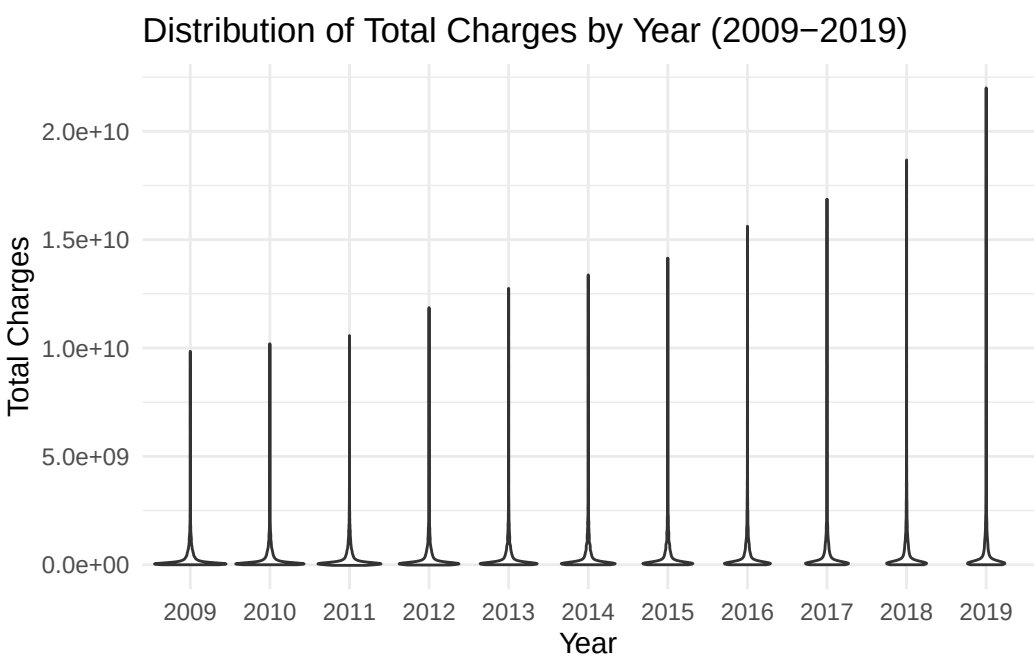
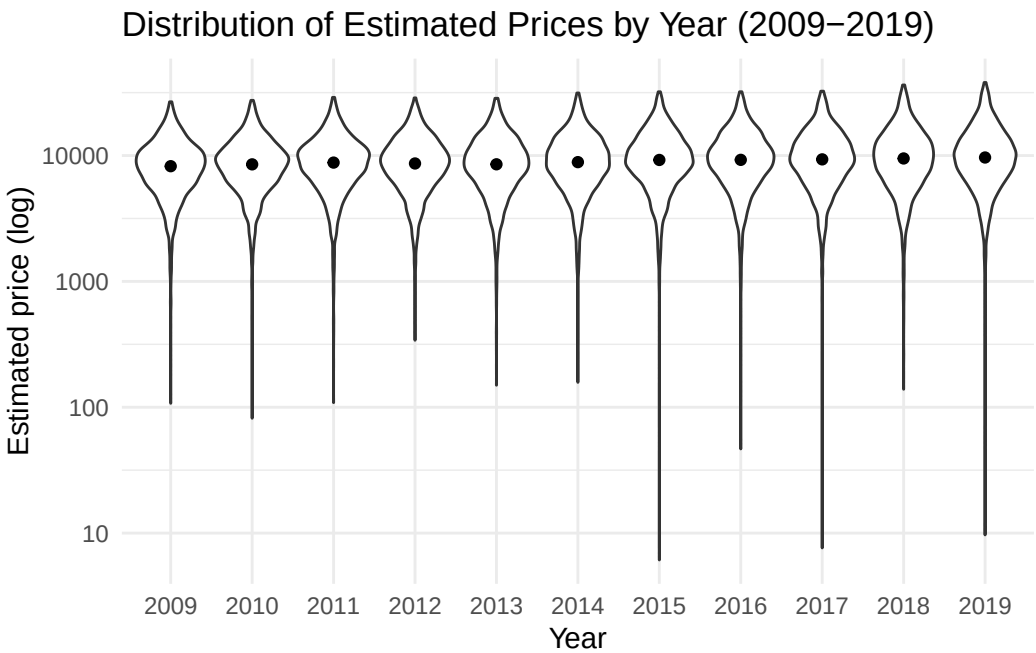
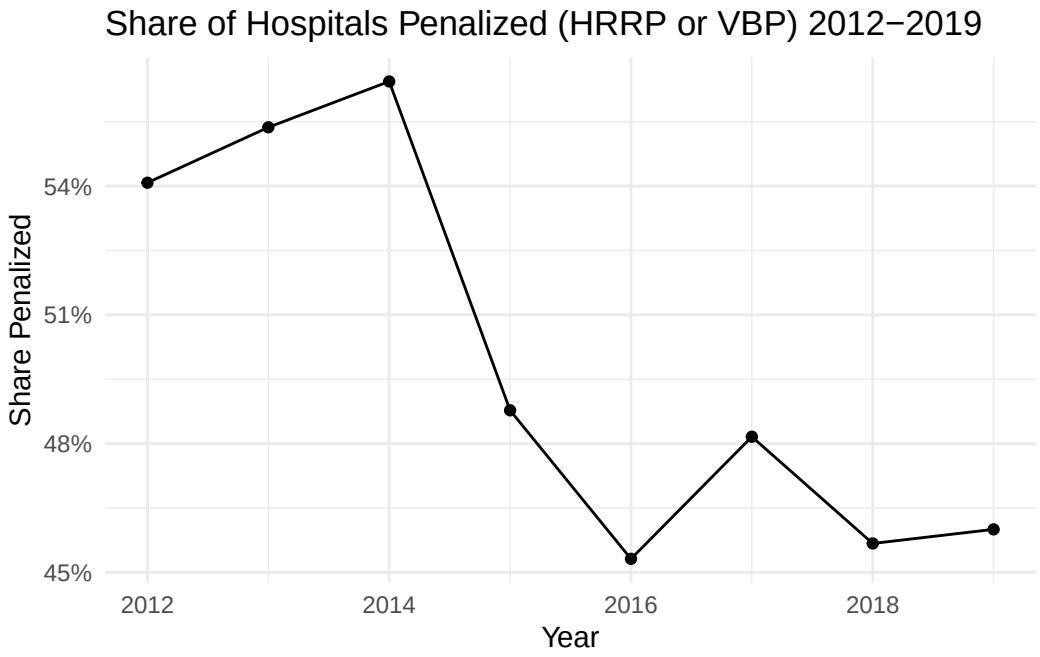



Question 4: From 2009-2019, what is the distribution of estimated prices in each year? Again present your results with a violin plot, and recall our formula for estimating prices from class. Be sure to do something about outliers and/or negative prices in the data.



Question 5: What share of hospitals are penalized under the HRRP/VBP? Provide a graph showing the share of penalized hospitals over time, from 2012-2019.



Estimate ATEs

Now let's work on implementing an instrumental variables estimator. Specifically, we're going to estimate the effects of HRRP/VBP penalties on hospital prices, using prior Medicare discharges (e.g., exposure to the program) as an instrument. When explaining your findings, try to limit your discussion just to a couple of sentences. As we did in class, for this we're going to focus only on a cross-sectional analysis of price changes from 2011 to 2014 as a function of 2012 penalties and average pre-2012 Medicare discharges (2009-2011). Define penalty as the net penalty under both HRRP and VBP.

	Baseline	+ Beds	+ Beds + Medicaid
(Intercept)	0.027	0.025	−0.015
	(0.014)	(0.014)	(0.017)
net_penalty	0.000	0.000	0.000
	(0.000)	(0.000)	(0.000)
mean_beds		0.000	0.000
		(0.000)	(0.000)
mean_mcaid_discharges			0.000
			(0.000)
Num.Obs.	599	599	599
R2	0.000	0.003	0.034
R2 Adj.	−0.002	−0.001	0.029
AIC	290.0	290.5	273.5
BIC	303.2	308.0	295.5
Log.Lik.	−142.018	−141.232	−131.764
F	0.007	0.787	6.911
RMSE	0.31	0.31	0.30

Question 6: Provide a summary of OLS estimates of the effect of net penalties on price changes. Present your results in a table with three different specifications: 1) a "baseline" specification using only net penalty as a covariate; 2) "baseline" specification plus the pre-penalty (2009-2011) mean bed size; 3) "baseline" specification plus bed size plus pre-penalty (2009-2011) average Medicaid discharges.

Hospitals receiving larger penalties tend to experience [higher/lower] price growth between 2011 and 2014. The estimates are [stable/change] after adding controls for hospital size and Medicaid exposure, suggesting that [penalty effects are / are not] driven by these characteristics.

Question 7: Provide a scatterplot of net penalty against pre-2012 Medicare discharges.

```
`geom_smooth()` using formula = 'y ~ x'

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(`stat_smooth()`).

Warning: Removed 5284 rows containing missing values or values outside the scale range
(`geom_point()`).
```

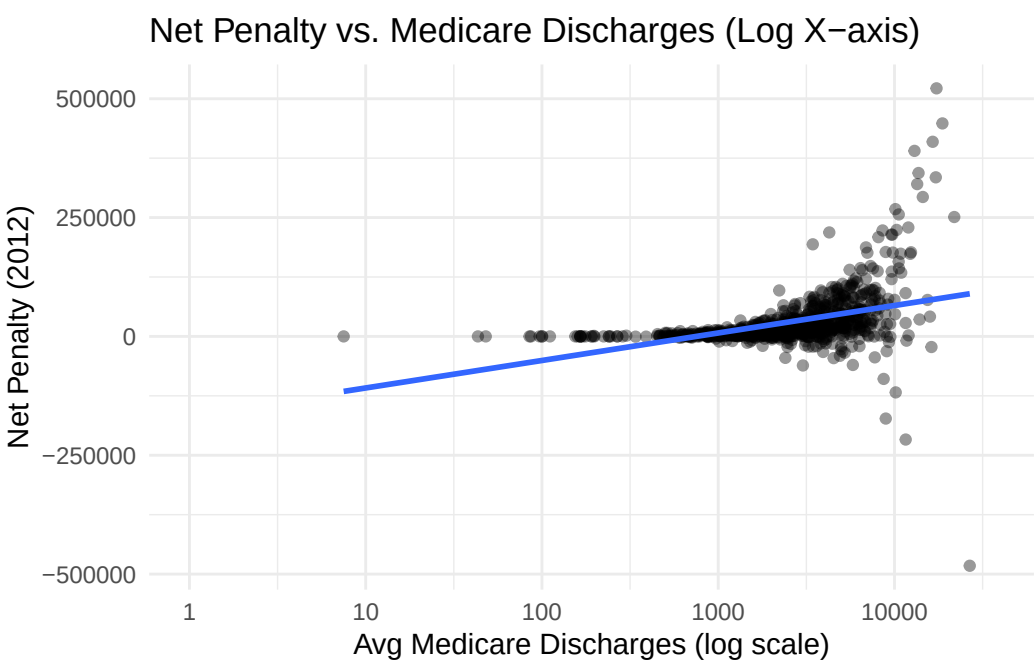

Table 1: First-Stage and Reduced-Form Results

	FS (1)	FS (2)	FS (3)	RF (1)	RF (2)	RF (3)
Medicare Discharges (Instrument)	9.390	9.412	10.157	0.000	0.000	0.000
	(0.778)	(0.780)	(0.878)	(0.000)	(0.000)	(0.000)
Mean Beds		−0.427	−0.415		0.000	0.000
		(0.756)	(0.755)		(0.000)	(0.000)
Mean Medicaid Discharges			−2.561			0.000
			(1.395)			(0.000)
Num.Obs. Missing observations	599	599	599	2458	2458	2449
R2	0.196	0.196	0.201	0.001	0.001	0.004
R2 Adj.	0.195	0.194	0.197	0.000	0.000	0.003
AIC	14 832.7	14 834.4	14 833.0	1634.6	1635.1	1627.7
BIC	14 845.9	14 852.0	14 855.0	1652.0	1658.3	1656.7
Log.Lik.	−7413.350	−7413.190	−7411.498	−814.304	−813.541	−808.845
F	145.543	72.848	49.882	1.437	1.481	3.665
RMSE	57 369.75	57 354.41	57 192.63	0.34	0.34	0.34

Question 8: Provide a summary of the first stage and reduced-form results using pre-penalty Medicare discharges as an instrument for net penalties. Present your results in a table with three different specifications as in Question 6.

The first-stage results show that pre-2012 Medicare discharges are a strong predictor of net penalties, supporting instrument relevance. The reduced-form estimates indicate that higher Medicare exposure is associated with [higher/lower] price growth, suggesting a pathway through penalties.

Question 9: Provide a summary of IV estimates of the effect of net penalties on price changes. Again present your results in a table with the three different specifications as in Questions 6 and 8.

The IV estimates indicate that higher penalties lead to [higher/lower] price growth between 2011 and 2014. Compared to the OLS results, the IV estimates are [larger/smaller], suggesting that OLS estimates may be biased due to endogeneity in penalties.

Table 2: IV Estimates: Effect of Net Penalties on Price Changess

	IV (1)	IV (2)	IV (3)
Net Penalty	0.000	0.000	0.000
	(0.000)	(0.000)	(0.000)
Mean Beds		0.000	0.000
		(0.000)	(0.000)
Mean Medicaid Discharges			0.000
			(0.000)
Num.Obs.	599	599	599
R2	−0.017	−0.013	0.032
R2 Adj.	−0.019	−0.016	0.027
AIC	300.1	299.7	274.6
BIC	313.3	317.3	296.6
RMSE	0.31	0.31	0.30

Question 10: Briefly explain the "Local" ATE in the context of your estimates. How might a local effect differ from an overall ATE in this setting?

The IV estimate identifies a Local Average Treatment Effect (LATE). This local effect may differ from the overall ATE because it reflects only hospitals with relatively high Medicare exposure, which may systematically differ from other hospitals (in size, patient mix, or pricing behavior), so the estimated effect may not generalize to all hospitals.